Digital Twin Backend and 3D Monitoring Platform

Cyber-Physical Structural Digital Twin for Cable-Stayed Bridges

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Abstract

An end-to-end digital twin that fuses **100 Hz vibration telemetry**, **online FFT modal analysis**, **explainable AI defect detection** (ResNet-18 + Grad-CAM), **2D-to-3D defect raycasting**, a **Three.js WebGL workstation** and an automated **AASHTO MBE** condition-index report, with a zero-latency standalone simulation fallback.

Problem: 5 SHM Challenges

- **Intermittent, subjective surveys** – manual, multi-year, hazardous
- **Isolated telemetry** – no real-time spectra or 3D link
- **Black-box CNNs** – no visual justification
- **No spatial grounding** – defects stay in 2D pixels
- **Slow compliance** – weeks of AASHTO paperwork

System Architecture

1 · Sensor Ingestion – tri-axial MEMS accelerometers (P1 base, mid-span, P2 base), 100 Hz, N = 1024 frames, WebSocket

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2 · Modal FFT – PSD, peak-picking f_0, Δf and ΔK

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3 · Vision & Saliency – ResNet-18 + Grad-CAM (crack, corrosion, sound)

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4 · 2D→3D Raycast – pixel centroid → [X, Y, Z] on FE mesh

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5 · AASHTO Reporting – BCI rating, A4 PDF export

Stack: FastAPI · PyTorch · SciPy · SQLite ledger · React 18 · Three.js r160 · Vercel

Dataset

12,450 annotated images (SDNET2018, CCIC, steel corrosion), 80/20 split (9,960 train / 2,490 val), 3 classes. Training: PyTorch 2.2, AdamW, RTX 4090.

Methodology

Vibration synthesis

$$a(t) = \Sigma\, A_k \sin(2\pi f_k t + \varphi_k)\, e^{-\zeta_k \omega_k t} + N(0, \sigma^2)$$

Spectral power & resonant peak

$$P(f_k) = 2|A[k]|^2 / (N \cdot f_s) \quad ; \quad f_0 = \arg\max P(f)$$

Stiffness loss from frequency shift

$$\Delta K / K_0 \approx 1 - (f_{obs} / f_{nom})^2$$

Grad-CAM saliency (layer4)

$$\alpha_k^c = (1/UV)\, \Sigma\Sigma\, \partial Y^c / \partial A^k_{ij}$$

$$L_{Grad\text{-}CAM} = \mathrm{ReLU}(\Sigma_k\, \alpha_k^c\, A^k)$$

2D→3D raycast projection

$$r = R^{-1}(K^{-1}[u_c\ v_c\ 1]^T)$$

$$P_{world} = C_{cam} + (-(n \cdot C_{cam} + d)/(n \cdot r))\, r$$

Bridge Condition Index

$$BCI = \Sigma\, w_i H_i / \Sigma\, w_i$$

Weights: main piers 2.0 · stay cables 1.5 · road decks 1.0 · railings 0.8

Decision Policy (AASHTO)

Trigger	Action / State
H < 50, Δf ≥ 0.35 Hz or ΔK ≥ 0.30	**Critical** – CS4, closure
H < 70, Δf ≥ 0.20 Hz or A_d ≥ 150 cm²	**Repair** – CS3
H < 85 or Δf ≥ 0.10 Hz	**Preventive** – CS2
Otherwise	**Monitor** – CS1

Component Health (AASHTO)

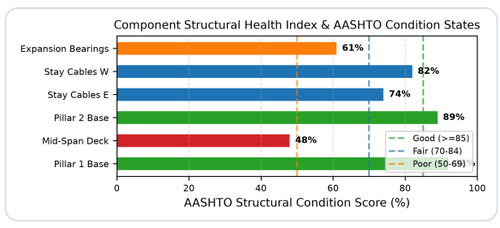


Fig. 2 – Component condition index and AASHTO states

Conclusion, Limits & Future

- Detects internal stiffness loss and explains visual defects, anchored in 3D
- **Limits:** synthetic accelerometer data; planar raycast parallax on curved cables; thermal compensation needed
- **Future:** UAV waypoints, FBG sensing, PINNs, regional twin, BIM/IFC sync
- **Generalizes to** wind turbines, dams, tunnels, aircraft, towers

Key Results

95.14% Accuracy	**0.834** Mean IoU
12.5 ms Inference latency	**<8.2 ms** Telemetry latency
60 FPS 3D rendering	**36.0%** Mid-span ΔK found

Model Benchmark

Model	Acc.	F1	AUC	Latency
VGG-16	92.45%	0.891	0.971	34.2 ms
MobileNet-V2	93.82%	0.908	0.981	8.5 ms
ViT-B/16	95.68%	0.934	0.992	58.7 ms
ResNet-18 (Ours)	**95.14%**	**0.928**	**0.989**	**12.5 ms**

ViT is marginally more accurate but 4.7× slower; ResNet-18 is the best trade-off for inline operation.

Grad-CAM keeps **88.4%** of saliency density inside actual crack boundaries. Crack IoU 0.824, corrosion 0.791, sound 0.887.

Modal Stiffness Detection

Sensor	Nominal	Observed	Δf	Status
S1 Pillar 1	3.20	3.20	0.00	Nominal
S2 Mid-span	**1.85**	**1.48**	**0.37**	**Warning (64% K)**
S3 Pillar 2	3.20	3.18	0.02	Nominal

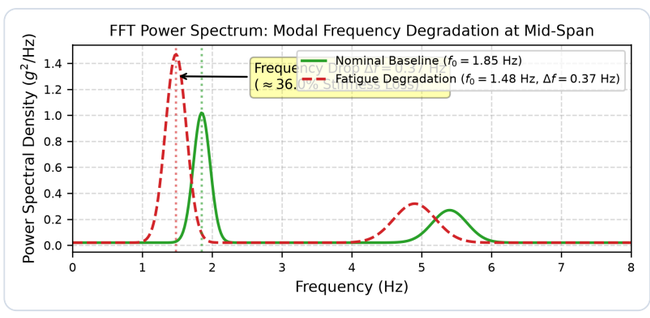


Fig. 4 – FFT PSD: mid-span peak shifts 1.85 → 1.48 Hz

References: He et al. (ResNet, CVPR 2016) · Selvaraju et al. (Grad-CAM, ICCV 2017) · Farrar & Worden (SHM, 2007) · AASHTO MBE 3rd Ed. (2018) · Cha et al. (CACAIE, 2017)